

Can large language models autonomously generate unique and profound insights in fundamental analysis?

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ABSTRACT

Fundamental analysis plays a critical role in equity investing, but its complexity has long limited the involvement of artificial intelligence (AI). Recent advances in large language models (LLMs), however, have opened new possibilities for AI to handle fundamental analysis. Despite this potential, leveraging LLMs to generate practically useful outputs remains a non-trivial challenge, and existing research is still in its early stages. This paper aims to enhance the performance of LLMs in fundamental analysis in a novel way, drawing inspiration from the practices of human analysts. We first propose a novel Autonomous Fundamental Analysis System (AutoFAS), which enables LLM agents to perform analyses on various topics of target companies. Next, we allow LLM agents to autonomously conduct research on specified companies with AutoFAS by exploring various topics they deem important, mimicking the experience accumulation of human analysts. Then, when presented with new research topics, the agents generate reports by referring to their accumulated analyses. Experiments show that, with AutoFAS, LLM agents can autonomously and logically explore various facets of target companies. The evaluation of their analysis on new research topics demonstrates that by drawing on accumulated analyses, they can naturally produce more unique and profound insights. This resembles the human process of generating novel ideas. Our work highlights a promising direction for applying LLMs in complex fundamental analysis, bridging the gap between human expertise and LLMs' analysis.

1. Introduction

Fundamental investing is a mainstream investment approach. Textual information is an important data source for this approach. Such information includes company disclosures, news articles, policy and regulatory documents, etc. In recent years, many researchers have explored the use of natural language processing (NLP) technology to process such textual information. Among these efforts, developments in stock price prediction and company analysis tasks are as follows.

Before the large language model (LLM) ChatGPT was released at the end of 2022, a popular approach was to utilize encoder models such as BERT (Devlin et al., 2019) to convert news text into vectors, which are then integrated with other data to feed into neural networks to predict stock prices (Cheng et al., 2022; Ding et al., 2015; Luo et al., 2023; Sawhney et al., 2021; Xu et al., 2021). Since the reasoning processes of encoder models and those neural networks are not interpretable to humans, it is often difficult to intuitively understand how the predictions are deduced.

With the significant progress of LLMs, their powerful capabilities and ability to generate understandable reasoning processes have garnered widespread attention. Several studies have investigated LLMs capabilities in some basic financial tasks including sentiment analysis, causal mining, mathematical reasoning and question answering (Dowling & Lucey, 2023; Li et al., 2023; Niszczoła & Abbas, 2023; Takayanagi et al., 2024; Xing, 2025; Yang et al., 2023; Zhao et al., 2024). For stock price prediction, Lopez-Lira and Tang (2024) use ChatGPT to judge the sentiment of news headlines and examine the power of ChatGPT predicting stock returns. Pelster and Val (2024) found a positive correlation between ChatGPT ratings based on financial news, and stock returns. Chiu et al. (2025) confirmed that ChatGPT can extract key information from financial news that reflects overall stock market movements. (Fatouros et al., 2024) and Papasotiriou et al. (2024) not only feed company news but also financial figures or macroeconomic information to GPT-4 for stock prediction. Further, Wang et al. (2024a) enhances LLM capabilities by instructing them to extract profitable factors from news before predicting stock prices. Yu et al. (2023)

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and Koa et al. (2024) employ fine-tuning to improve LLM stock price predictions based on financial news.

The aforementioned studies primarily use news for stock price prediction. However, news generally do not cover all specific details and lack information about a company's fundamentals. While signals for quantitative investing may be extracted from news, relying solely on news is often insufficient for making investment decisions in fundamental investing. In practice, the primary information source for fundamental investing is company releases, which include not only quantitative financial data but also qualitative descriptions of company performance. Fundamental analysts primarily base their analyses on these releases and only perform in-depth analyses of news in conjunction with company fundamentals when significant news is released.

Before ChatGPT was released, NLP and other machine learning techniques had difficulty handling all the complicated workflows of fundamental analysis. Fundamental investing involves extensive analysis of both structured numerical data and unstructured text data, covering a wide range of factors from individual company conditions to broader economic trends (Valentine, 2011). For a long time, human fundamental analysts have had to devote a significant amount of effort to carrying out the analysis. Due to their powerful capabilities and generative nature, LLMs have unlocked new possibilities for AI to assist humans in fundamental analysis.

Recent studies on using LLMs for fundamental analysis based on company releases mainly include the following: Xu et al. (2025) employ LLMs to generate analysis reports based on company earnings releases. They enhance the analysis capability of LLMs by introducing a multi-agent framework to simulate the analysis process of human analysts. An LLM agent is a system designed to autonomously execute tasks by leveraging LLMs (Wang et al., 2024). Takayanagi et al. (2025) use GPT-4 to generate analysis reports from earnings call transcripts. Their findings indicate that while these reports can influence investors' decisions, there remains a noticeable gap in usefulness compared to reports by human analysts. Goldsack et al. (2025) employ GPT-4 to generate analysis reports based on earnings call transcripts, accompanying phonetic statistics and financial data. They incorporate a multi-agent framework to iteratively enhance the reports. Fatouros et al. (2025) introduce a multi-agent framework that analyzes diverse financial information sources including news articles, earnings releases, price dynamics, and macroeconomic indicators, and integrates their outputs to generate investment signals.

From the above, we can see that research on utilizing LLMs for fundamental analysis is still in its early stages. Although some studies (Fatouros et al., 2025; Goldsack et al., 2025; Koa et al., 2024; Xu et al., 2025) have incorporated specific frameworks or finetuning to enhance the analytical capabilities of LLMs, existing studies on fundamental analysis and stock price prediction typically require models to generate analysis directly from raw company disclosures or news texts. In light of the practices of human analysts, the lack of prior analytical accumulation would make it difficult for LLMs to produce high-quality fundamental analysis.

In practice, experienced fundamental analysts accumulate a wealth of knowledge about the companies in their coverage through years of research. For instance, they may uncover important trends by calculating financial ratios, identify key performance drivers through long-term tracking, or distinguish unique characteristics by comparing companies within the same sector. Such knowledge and insights are rarely directly stated in company disclosures and instead must be inferred through careful analysis. As a result, when conducting new analyses, analysts can draw upon their prior research to provide unique and profound insights, which in turn help identify more investment opportunities and potential risks.

Building on the above observation, we propose a novel direction to improve the quality of fundamental analysis generated by LLMs. First, we enable LLM agents to autonomously conduct comprehensive

analyses of companies within a specific sector. Once a certain number of analyses have been accumulated, LLM agents are then tasked with analyzing specific topics while referring to these accumulated analyses. Our study addresses a gap in previous research by introducing this prior accumulation approach.

To put our idea into practice, we require a system that allows LLM agents to autonomously conduct comprehensive analyses. To date, research on utilizing LLMs for fundamental analysis and stock price prediction has typically involved designing systems that analyze a single company at a time based on pre-specified information for a specific theme (earnings analysis or stock price prediction). However, in practice, analysts selectively employ relevant information to analyze various research topics for multiple companies within the same sector. Based on this characteristic, we developed a novel Autonomous Fundamental Analysis System (AutoFAS) that enables LLM agents to:

1. Analyze a wide range of research topics.
2. Independently extract relevant information from company releases according to research needs.
3. Simultaneously analyze multiple companies within the same sector, thereby deepening their understanding of each company from a sector-wide perspective.

Within this system, we conducted the following studies to implement and validate our idea:

1. First, we enabled LLM agents to autonomously conduct comprehensive analyses of companies within a specific sector, while recording the analyses generated during the process. We refer to this process as the self-development of agents. Agents that have undergone self-development are termed Expert Agents, while those that have not are termed Novice Agents.
2. We assign several research topics to both the Expert Agent and the Novice Agent, instructing them to generate analysis reports. We then evaluate these reports to determine whether the Expert Agent produces more unique and profound insights.

Our experiments validate our idea. Much like human analysts, after LLM agents conduct comprehensive analyses on a specific sector, they naturally generate higher-quality reports based on the accumulated analyses. Specifically, the reports produced by Expert Agents provide more analyses on long-term company strategies and forecasts, and offer more unique and profound insights. The human analyst evaluation in Table 5 shows that the frequency of Expert Agents' reports that contain more long-term factors is 17%, compared to 8% for Novice Agents; unique perspectives are 35% versus 4%; and profound insights are 24% versus 3%.

The objectives of this study are twofold. First, we aim to develop a system that enables LLM agents to autonomously conduct company analyses by selecting important research topics and extracting necessary information. Second, we aim to enhance the quality of fundamental analysis on new research topics by enabling LLM agents to accumulate company-specific knowledge through such autonomous analysis.

Our contributions are:

1. We developed the system AutoFAS and empirically demonstrated that within the system, LLM agents can autonomously and logically perform in-depth analyses of various aspects of target companies. Moreover, we show that once a sufficient number of analyses have been accumulated, the agents are able to generate more unique and profound insights on new research topics.
2. Our study reveals a novel and promising direction for enhancing the performance of LLMs in fundamental analysis. While existing studies have primarily focused on improving LLMs' ability to derive conclusions directly from raw textual information, we

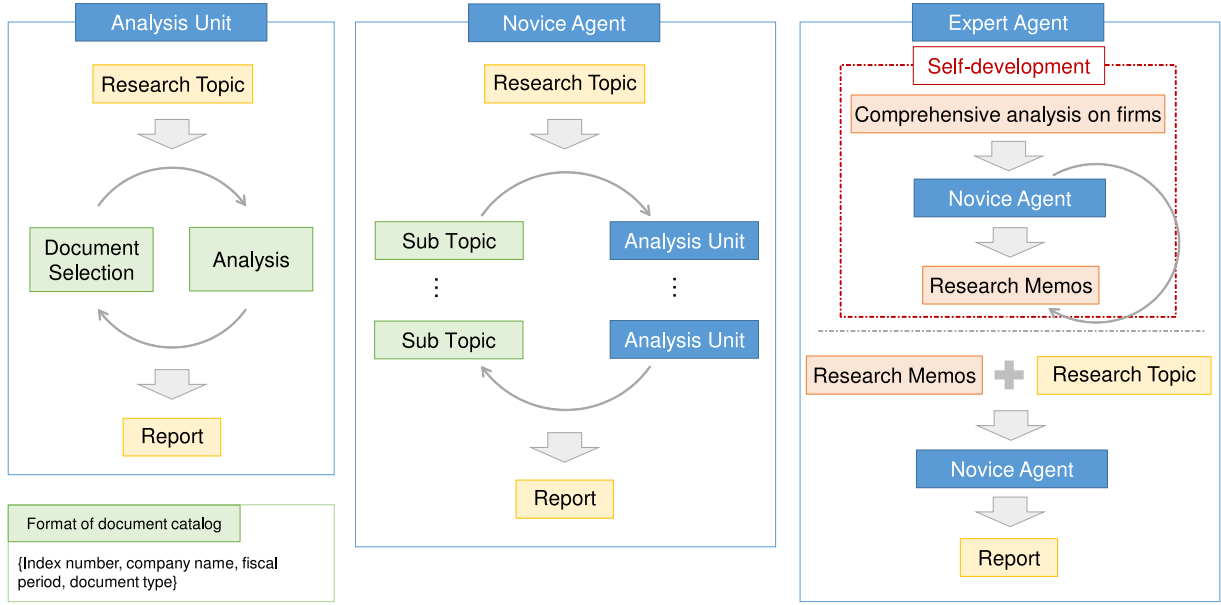


Fig. 1. Design of our Autonomous Fundamental Analysis System. The Analysis Unit is an agent responsible for analyzing an assigned topic. It selects the necessary documents from document catalog to conduct its analysis, and if further documents are required, it continues to select what it needs, repeating this cycle to complete the analysis. The Novice Agent is an agent tasked with performing comprehensive analysis of a given topic. It considers from which aspects the assigned topic can be analyzed and pass instructions to Analysis Units for each sub topic. After receiving responses from the Analysis Units, it further considers which additional aspects need to be analyzed. Through this iterative cycle, the Novice Agent completes a deeper analysis. The Expert Agent is an agent that has accumulated a certain amount of analysis by instructing Novice Agents to conduct comprehensive analysis of various aspects of companies within a sector. When the Expert Agent is assigned a topic to analyze, it refers to the accumulated analyses. The difference between the Expert Agent and the Novice Agent is whether or not they hold accumulated analyses.

proposed and validated a novel approach that enables LLMs to perform analyses after accumulating prior research. This study provides new insights into how LLMs can be better adapted for fundamental analysis tasks.

The remainder of this paper is organized as follows: Section 2 explains our proposed method. Section 3 details the experimental settings. Section 4 presents the results and discussion. Finally, Section 5 concludes the paper.

2. Methodology

This section describes the design of AutoFAS and the evaluation method for reports generated by Novice and Expert Agents.

2.1. System design

Fig. 1 illustrates the components of our proposed AutoFAS. First, we build an LLM agent called the “Analysis Unit”, whose task is to analyze a specified research topic. Next, to achieve more comprehensive analysis, we introduce the “Novice Agent”. Upon receiving a research topic, it identifies aspects to explore and distributes sub-topics to the Analysis Units. Based on the results, it reassesses and assigns new directions for analysis. This cycle continues until the Novice Agent determines the topic has been thoroughly analyzed. To enable the Novice Agent to develop into an Expert Agent for a specific sector, we instruct it to carry out a comprehensive analysis of companies in that sector, iterating the analyses until reaching a predetermined number of iterations. The specific design of each agent is detailed as follows.

Analysis Unit: The Analysis Unit can perform the following three actions:

1. **Document Selection:** Select documents that are relevant to the current research topic from a database.
2. **Analysis:** Conduct analysis based on the selected documents.
3. **Completion:** Conclude the analysis process and generate the final report.

As shown in the prompt examples in Table 1, the agent is instructed to choose the most appropriate action from these three at each step. Upon receiving a research topic, the unit first selects relevant documents, then conducts the analysis; if further exploration is deemed necessary, it repeats this cycle until it decides to conclude the analysis.

Regarding “Document Selection”, in our experiments, we built a database containing three types of quarterly company earnings data: key financial indicators in financial statements, financial reports that include qualitative explanations on business performance, and earnings call transcripts. Considering that LLMs are required to analyze multiple companies within the same sector using both current and historical documents, directly inputting all documents at once would exceed the token limit. On the other hand, employing a Retrieval-Augmented Generation (Guu et al., 2020) method based on keyword extraction might result in information loss (Li et al., 2025). Based on these considerations and drawing inspiration from the information retrieval practice employed by human analysts, we designed the following mechanism:

We first create a catalog of all documents related to the target companies. This catalog records key information for each document in the following format: {Index number, company name, fiscal period, document type}. The catalog is then provided to the agent, which selects the index numbers of the documents it deems necessary. The full text of the selected documents is subsequently fed into the agent through prompt for analysis. If, during the analysis, the agent determines that additional documents are needed, the text of the previously used documents are cleared from prompt before feeding the new document texts in order to avoid exceeding the token limit.

Novice Agent: For a given research topic, the Novice Agent conducts a comprehensive analysis from multiple perspectives. At each step, the agent selects one of the following three actions to execute:

1. **Thinking:** Review the analysis completed so far and consider which additional angles to explore.
2. **Task Assignment:** Assign specific sub topics to the each Analysis Unit.
3. **Completion:** Conclude the analysis process and generate the final report.

Table 1
Prompts for the Analysis Unit and the Novice Agent.

Analysis Unit	Novice Agent
You are a senior fundamental analyst working at a financial institution. At each step, select and perform only one of the following actions: 1. Document Selection: Select documents that are relevant to the current research topic from a database. 2. Analysis: Conduct an analysis based on the selected documents. 3. Completion: Conclude the analysis process and generate the final report. Below is the catalog of available documents: {documents catalog} Your instruction is: {research topic}	You are a senior fundamental analyst working at a financial institution. You are covering {cover company names}. You have team members to whom you can give analysis instructions. Generated analyses are recorded in as research memos. By repeatedly performing the following actions, you progressively deepen your analysis of the covered companies. At each step, select and perform only one of the following actions: 1. Thinking: Review the analysis completed so far and consider which additional angles to explore. 2. Task Assignment: Assign specific sub topics to the each Analysis Unit. 3. Completion: Conclude the analysis process and generate the final report. Your instruction is: {research topic}

The Novice Agent alternates between the “Thinking” and “Task Assignment” actions, gradually deepening its understanding of the research topic until it deems the analysis sufficiently thorough, at which point it selects the “Completion” to generate the final report. The prompt for this agent is in [Table 1](#).

Expert Agent: We instruct the Novice Agent to conduct a comprehensive analysis of companies in each specific sector, starting with reviewing financial results. By repeatedly executing the “Thinking” and “Task Assignment” actions, the agent gradually analyzes various aspects of companies within that sector. All the analysis produced during this process is stored as text in research memos. We refer to this process as the agent’s self-development. Once the Novice Agent has accumulated a sufficient amount of analysis as research memos, we designate it as an Expert Agent. When the Expert Agent analyzes a new research topic, the research memos are included in the prompt fed to the agent. Thus, an Expert Agent is equivalent to a Novice Agent that holds research memos.

2.2. Evaluation method

To assess the impact of the accumulated analysis by the Expert Agent on its report generation, we assigned a series of research topics to both the Novice Agent and the Expert Agent, instructing each to generate analysis reports. We then evaluate the report pairs based on the eight criteria listed in [Table 2](#) to determine which agent performed better. The evaluation is conducted in a relative manner, focusing on determining which of the two reports performs better in each criterion. Among these criteria, Criteria 1.1–2.2 primarily assess the basic features of the reports, while Criteria 3.1–4.2 focus on whether the reports offer unique perspectives and profound insights.

We employed a hybrid evaluation strategy, combining multiple rounds of comprehensive assessments by LLMs with evaluations of randomly selected report pairs by three professional fundamental analysts. We consider an evaluation reliable if both parties agree on which agent performs better on a given evaluation criterion. [Goldsack et al. \(2025\)](#) demonstrates that LLMs largely achieve good alignment with human experts in the evaluation of financial analytical reports. However, in our experiment, we include professional analysts to mitigate the influence of any inaccuracies in LLM evaluations, while involving a few professionals to assess part of the data is due to the practical challenges of coordinating a large number of professionals for a full-scale evaluation. Although non-experts could be used for evaluation, empirical studies indicate that in the highly specialized field of finance, non-expert assessments are often insufficiently accurate ([Takayanagi et al., 2025](#)).

3. Experiment settings

We selected five major sectors in the Japanese equity market: Auto, Steel, Trading Company, Telecommunications, and Insurance. For each sector, we chose three major companies. The data used in the experiment consists of quarterly key financial indicators, financial reports and earnings call transcripts from January 2022 to June 2024. The financial indicators and financial reports were obtained from the companies’ official websites, while the earnings call transcripts were sourced from QUICK Corp. We use GPT-4o as the underlying model for all agents. For each sector, we instructed a Novice Agent to perform a total of 50 rounds of “Thinking” or “Task Assignment” actions. The generated analyses along the way are recorded as research memos.

The research topics used in the evaluation of Expert Agents were generated as follows: First, we fed the 2024 financial reports and earnings call transcripts to GPT-4o. Then, we instruct it to generate 5 research topics for each company, along with 5 topics for inter-company comparisons. This yields 20 research topics per sector, resulting in a total of 100 topics across five sectors. Some examples of the research topics are as follows: “Comparison of mid-term business plans and growth strategies of each company”, “Analysis of how the growth strategy in non-resource sectors has contributed to profit”, and “Analysis on the impact of new KPIs on long-term shareholder return policies”.

For each research topic, the Novice Agent and the Expert Agent each generate a report. We then have GPT-4.1 compare the report pairs six times independently, corresponding to six evaluators. To eliminate any bias caused by the order in which the reports were presented, each report pair was evaluated three times in one order and three times in the reverse order. For evaluation by three professional fundamental analysts, we randomly select reports corresponding to 10 research topics from each sector, resulting in a total of 50 topics.

4. Results and discussion

This section presents the experimental results, including an examination of the types of analyses generated during the self-development process and a comparison of the reports produced by Novice and Expert Agents. We further investigate how accumulated analyses affect the generation of new reports, examine the impact of varying levels of research accumulation, assess the effect of including research memos in the prompts for LLM evaluation, and evaluate the impact of different prompts on report generation.

Table 2
Evaluation criteria.

1.1 Quantitative Analysis	Which report incorporates more detailed numerical data in its analysis?
1.2 Qualitative Analysis	Which report conducts a more extensive qualitative analysis?
2.1 Short-Term Factors	Which report provides a more detailed explanation of past performance and short-term conditions?
2.2 Long-Term Factors	Which report offers a more detailed discussion of long-term strategies and outlook?
3.1 Direct Relevance	Which report presents more detailed elements directly related to the research topic?
3.2 Unique Perspectives	Which report, beyond directly relevant elements, introduces unique perspectives?
4.1 Basic Information	Which report provides more detailed descriptions of the basic information about the company?
4.2 Profound Insights	Which report goes beyond basic company information and provides profound insights?

4.1. Results of self-development

First, we examine how and what analyses have been conducted by Expert Agents during their self-development.

Table 3 presents the thinking processes of Expert Agent for the five sectors. The thinking processes are obtained by asking the Expert Agent how it decided what to analyze during the self-development process after its completion. We can observe that the agent gradually deepen and expand its analytical focus. Taking auto sector for example, by reviewing financial results, the agent identified several key factors influencing company performance. It then delved into these factors to assess their impact on financial performance. Subsequently, it conducted a detailed analysis at the segment and regional levels before evaluating the long-term growth prospects of these companies. Last, as technological innovation was recognized as a core element of long-term strategy, the agent assessed its impact on profitability. This process is similar to human.

Table 4 presents the number of analyses generated, and the distribution of topics across the five sectors. The number of memos generated during self-development ranges from 70 to 75 across sectors. We categorize these memos into several groups for each sector, with the percentage next to each category indicating the proportion of memos belonging to that category. Basic categories such as financial analysis, shareholder returns, growth strategy, and segment performance are commonly examined across all sectors. Additionally, sector-specific topics are also addressed. For instance, electrification strategy in the auto sector and technological innovation in the steel sector.

Thus, the self-development process of Expert Agents follows a logically coherent progression, producing well-structured analyses across various aspects of each sector.

4.2. Evaluation of reports generated by novice agents and expert agents

In this section, we examine the impact of the accumulated analysis of the Expert Agent on the new reports it generates. Table 5 presents the evaluation results by human analysts and GPT-4.1. The percentages indicate how often each report, generated by the Novice Agent and the Expert Agent respectively, was judged as superior to the other or of equivalent quality for each evaluation criterion. The results for each sector show only GPT-4.1's evaluations.

Within the eight evaluation criteria, “long-term factors”, “unique perspectives”, and “profound insights” are the three where the differences between the Novice and Expert Agents are significant and aligned in both human and GPT-4.1 evaluations. This aligns with our expectation that experienced analysts, through years of research accumulation, can offer more unique and insightful perspectives. This pattern is also evident in sector level evaluations by GPT-4.1, with the only exception of “long-term factors” and “unique perspectives” in the Trading Company sector and “long-term factors” and “profound insights” in the Telecommunications sector. Conversely, while the differences are less pronounced, the Novice Agent tends to provide more detailed descriptions in “short-term analysis” and “basic information”. For the remaining criteria, “quantitative analysis”, “qualitative analysis” and “direct relevance”, no consistent differences are observed.

These findings suggest that, after conducting comprehensive analysis of coverage companies, Expert Agents can generate research reports

that place greater emphasis on long-term factors, offer unique perspectives and profound insights. In the following section, we examine why Expert Agents possess these abilities.

4.3. How prior accumulation affects new generations

Does the above mentioned abilities of Expert Agents stem from the content accumulated in the research memos? To explore this question, we extracted reports where human analysts rated the Expert Agent as superior in these three criteria and conducted a detailed analysis of these reports.

First, we asked human analysts to highlight superior arguments in the Expert Agent's reports for these three criteria. Then, we examined the text generated by the Expert Agent during the “Thinking” actions to identify which research memos were referred. Then we verified whether these memos contained the highlighted arguments. The results show that in 81% of the reports, the arguments highlighted by analysts are found in the research memos referred by the Expert Agent. Table 6 provides an example. For the research topic “Analysis of the Global Market Strategy”, the Novice Agent focused on factual descriptions of each market. In contrast, the Expert Agent offered a deeper perspective by incorporating electrification strategies and offering insights on their importance for emerging markets at the conclusion. We confirmed that the research memos referred by the Expert Agent indeed include analyses related to electrification. Table 7 presents some other examples of research topics and referred research memos by Expert Agents. We can confirm that the Expert Agents extract relevant analyses from the accumulated memos, covering different aspects related to the research topics.

Although we did not require the agent to generate analyses on long-term factors or provide more unique and profound insights during its self-development, the above investigation shows that the Expert Agent can draw inspiration from the analyses in the research memos when conducting new research, thereby naturally producing higher-quality analyses. This process closely resembles human thought processes. According to the associative theory of creativity (Mednick, 1962), creativity involves the connection of weakly related, remote concepts into novel and applicable concepts. Empirical studies show that the more potentially relevant elements retrieved during thinking, the greater the likelihood of generating unusual associations and novel ideas (Amabile et al., 2005; Kenett, 2025; Ov et al., 2022). Our research indicates that LLMs also possess this characteristic.

We believe that this characteristic of LLMs stems from their “in-context learning” ability. This ability refers to the LLM's capacity to learn how to better perform tasks by providing a few examples within the prompt (Brown et al., 2020). For instance, in question-answering tasks, including a few examples of questions and their answers in the prompt can improve the accuracy of the LLM's responses. In our study, when the Expert Agent analyzes new research topics, the analyses recorded in the research memos serve as example analyses for these new topics. The Expert Agent can learn relevant perspectives from these examples and apply them to the analysis of new research topics, thereby producing more unique and profound insights.

Table 3
Thinking process of expert agent.

	Auto	Steel	Trading company
Thinking process	1. Review of Financial Results Reviewed the financial results of each company and summarized their focus on electrification initiatives, market strategies, and financial stability measures. 2. Analysis of Key Strategies and Financial Performance Based on the key areas identified in the financial results, conducted a quantitative assessment of the impact of electrification efforts, pricing strategies, and regional sales approaches on financial performance. 3. Evaluation of Competitive Environment Analyzed risks and opportunities in different regions and segments to identify factors that could accelerate growth. 4. Forecasting the Long-term Impact of Business Strategies Conducted analysis to assess the long-term impact of business strategies on corporate value. 5. Cost-effectiveness of Technological Innovation As a core element of business strategy, technological development is quantitatively assessed to evaluate its impact on profitability.	1. Review of Financial Results Reviewed the financial results of each company to understand the overall business performance, key financial indicators, and changes in management policies. 2. Analysis of Strategy and Market Expansion Based on the management policies identified in the financial results, each company's overseas expansion and growth strategies were examined. 3. Assessment of Risk Management and Financial Soundness Considering each company's market expansion strategy, the associated risks and financial soundness were assessed. 4. Formulation of a Sustainable Growth Strategy The strategic decision-making required to sustain long-term growth was examined. 5. Evaluation of Contribution of Technological Innovation to Market Competitiveness In the context of growth strategy development, the impact of technological innovation on market expansion was assessed.	1. Review of Financial Results Reviewed the financial results of each company, including revenue, net profit, and other key financial indicators. 2. Identification of Key Analytical Themes Based on the analysis of financial results, key issues requiring further examination were identified. 3. Impact of Specific Strategies on Corporate Financials After clarifying the key analytical themes, the impact of specific strategies on corporate financial performance was analyzed. 4. Analysis of Competitive Environment and Market Trends The competitive environment and market trends were examined to understand how corporate financial performance and strategic initiatives are positioned within the broader competitive landscape.
	Telecommunications	Insurance	
Thinking process	1. Review of Financial Results: Reviewed the financial results of each company, identifying each company's overall performance and growth trends. 2. Analysis of Technological Innovation and Market Expansion: Analyzed each company's technological innovations and market expansion strategies to determine their contribution to performance. 3. Assessment of Strategic Initiatives and Portfolio: Assessed the effectiveness of each company's strategic initiatives and their business portfolios. 4. Risk Management and Sustainable Growth: Analyzed the risks faced by each company and their management strategies. 5. Growth Strategies: Conducted further analysis on growth strategies.	1. Review of Financial Results: Reviewed the financial results of each company, identifying the key factors they prioritized. 2. In-depth Analysis of Performance Impact: A comparative analysis was conducted to highlight differences between companies. 3. Evaluation of Competitive Environment: Further analyzed the competitive landscape. 4. Impact of Technological Innovation and Market Expansion: Next, an analysis was conducted to determine how the adoption of new technologies and market expansion efforts influenced competitiveness. 5. Review of Market Share Expansion Strategies: Finally, a review of each company's market share expansion strategies was conducted.	

Table 4
Statistics of research memos generated by Expert Agents during self-development.

	Auto	Steel	Trading company
Num of memos	75	72	70
Topics of memos	1. Financial Analysis and Shareholder Returns: 27% 2. Regional Strategies and Performance: 22% 3. Electrification Strategy: 24% 4. Technological Innovation: 15% 5. Strategic Partnerships: 12%	1. Financial Analysis: 22% 2. Financial Strategy and Risk Management: 15% 3. Business Strategy: 23% 4. New Investments: 18% 5. Technological Innovation: 11% 6. Social Responsibility: 11%	1. Financial Analysis and Shareholder Returns: 35% 2. Growth Strategy and Outlook: 33% 3. Segment Performance: 11% 4. Commodity Price: 15% 5. Sustainability: 6%
	Telecommunications	Insurance	
Num of memos	73	75	
Topics of memos	1. Financial Analysis: 34% 2. Regional Strategies: 16% 3. Growth Strategy and Sustainability: 12% 4. Introduction of New Technologies: 24% 5. Organizational Culture: 14%	1. Financial Analysis and Shareholder Returns: 48% 2. Underwriting Profit and Market Strategy: 9% 3. Overseas Business: 13% 4. Strategic Investments: 16% 5. Risk Management: 14%	

Table 5

Performance comparison of reports generated by Novice Agents and Expert Agents. Evaluated by three human analysts for 50 topics and six LLM evaluators for all the 100 topics.

	Analyst (50 topics)			GPT-4.1 (all 100 topics)			Auto			Steel		
	Novice	Expert	Equal	Novice	Expert	Equal	Novice	Expert	Equal	Novice	Expert	Equal
1.1 Quantitative Analysis	13%	21%	66%	54%	46%	0%	51%	49%	0%	59%	41%	0%
1.2 Qualitative Analysis	25%	21%	54%	47%	54%	0%	50%	50%	0%	45%	55%	0%
2.1 Short-Term Factors	10%	8%	82%	52%	49%	0%	60%	40%	0%	50%	50%	0%
2.2 Long-Term Factors	8%	17%	75%	48%	51%	1%	41%	58%	1%	37%	63%	0%
3.1 Direct Relevance	23%	9%	68%	48%	52%	0%	45%	54%	1%	56%	44%	0%
3.2 Unique Perspectives	4%	35%	61%	40%	60%	0%	37%	63%	0%	39%	61%	0%
4.1 Basic Information	17%	6%	77%	51%	49%	0%	59%	41%	0%	43%	57%	0%
4.2 Profound Insights	3%	24%	73%	46%	54%	0%	43%	56%	1%	43%	58%	0%
	Trading Company			Telecommunications			Insurance					
	Novice	Expert	Equal	Novice	Expert	Equal	Novice	Expert	Equal			
1.1 Quantitative Analysis	63%	38%	0%	55%	45%	0%	43%	58%	0%			
1.2 Qualitative Analysis	45%	55%	0%	44%	56%	0%	48%	52%	0%			
2.1 Short-Term Factors	63%	38%	0%	46%	54%	0%	39%	61%	0%			
2.2 Long-Term Factors	53%	48%	0%	62%	36%	3%	48%	53%	0%			
3.1 Direct Relevance	52%	48%	0%	49%	51%	0%	40%	60%	0%			
3.2 Unique Perspectives	52%	48%	0%	43%	58%	0%	31%	69%	0%			
4.1 Basic Information	59%	41%	0%	53%	47%	1%	40%	60%	0%			
4.2 Profound Insights	47%	53%	0%	51%	49%	0%	47%	53%	0%			

The percentages indicate how frequently each report, generated by the Novice Agent and the Expert Agent respectively, was judged as superior to the other or of equivalent quality for each evaluation criterion. For instance, the first figure, 13%, indicates that, on average across the three analyst evaluators, the reports generated by the Novice Agent were assessed as superior in 6.5 out of the 50 report pairs for the criterion of Quantitative Analysis. The results for each sector show only GPT-4.1's evaluations. Expert Agents excel in "long-term factors", "unique perspectives", and "profound insights".

Table 6

An example of reports generated by the Novice Agent and Expert Agent, along with the research memos referred by the Expert Agent.

Novice Agent	Expert Agent
1. Changes in Global Sales Volume (Omitted)	1. Sales Performance by Region
2. Sales Volume by Region	1.1 Japanese Domestic Market (Omitted)
2.1 Japanese Market (Omitted)	1.2 North American Market (Omitted)
2.2 North American Market (Omitted)	1.3 European Market (Omitted)
2.3 European Market (Omitted)	1.4 Emerging Market
2.4 Emerging Market	- Sales Volume: XX units (XX% decrease year-on-year).
- Sales Volume: XX units	Expanding market share amid intensified competition remains a challenge.
- Year-on-Year Change: XX% decrease	A shift toward new energy vehicles necessitates a reassessment of technology and product offerings to align with market demand.
The decline in sales was driven by intense competition in the emerging market. However, in the fourth quarter, sales increased by XX% compared to the same period of the previous year, showing signs of partial recovery.	2. Electrification Strategy
3. Conclusion	2.1 Impact and Results
The global market strategy has achieved notable success, particularly in the North American market. However, a strategic review is required to overcome challenges in the Emerging market.	- Global Electrified Vehicle Sales Ratio: Reached XX%, with significant progress in electrification.
Swift responses to fierce competition and securing sustainable growth remain crucial.	- Financial Performance: Revenue of XX trillion yen (XX% increase year-on-year), operating profit of XX billion yen (XX% increase year-on-year). The advancement of electrification technology has notably contributed to improved profit margins.
	2.2 Future Plans (Omitted)
	3. Conclusion
	While the growing demand for electrified vehicles in North America and Europe is driving revenue growth, challenges persist in the Emerging market. Future growth will require a strategic approach that capitalizes on the success of electrification while seizing expansion opportunities in emerging markets.
	===
	Titles of research memos referred:
	- Profit contribution from electrification strategy in the European market
	- Strategic positioning analysis in the European and North American markets
	- Strengthening of competitive strategy in the Chinese market
	- Strategic approach and effectiveness measurement in emerging markets
	- Quantitative analysis of promotional effects from the introduction of EV models
	- Impact of technological innovation on competitiveness

The research topic is "Analysis of the global market strategy". We identified the research memos referred during the analysis by looking into the "Thinking" process of the Expert Agent. We masked firm-specific numbers as "XX" since this paper is not intended to recommend any stock.

Table 7
Examples of research topics and referred memos by Expert Agents.

Sector	Research topic	Referred memos
Auto	Evaluation of the changes in sales by region and the future growth strategy	- Analysis of profit contribution from electrification strategy - Strategic positioning analysis in the European and North American markets - Strengthening of competitive strategy in the Chinese market - Strategic approach and effectiveness measurement in emerging markets - Quantitative analysis of promotional effects from the introduction of EV models - Impact of technological innovation on competitiveness
Steel	Comparison of mid-term business plans and growth strategies of each company	- Analysis of capital structure and capital utilization efficiency - Quantitative impact of technological innovations supporting competitive advantage - Improvement of competitiveness through technological innovation and market expansion strategies - Strategic positioning comparison among companies - Impact of each company's technological innovation and environmental considerations on their business model
Trading Company	Analysis of how the growth strategy in non-resource sectors has contributed to profit	- Detailed analysis of growth investments - ROI evaluation of growth investments - Profitability comparison by segment - Evaluation of strengthening measures in non-resource sectors
Telecom	Comparison of each company's cash flow utilization and investment plans, and evaluation of which company is strategically investing for future growth	- Formulation of specific tactics for market expansion - Evaluation of the sustainability of long-term strategies - Impact of next-generation technologies on cash flow - Evaluation of strategies aimed at improving profitability - Maintaining competitive advantage and leveraging new technologies
Insurance	Analysis on the impact of new KPIs and changes in dividend payout ratio on long-term shareholder return policies	- Long-term trends in capital efficiency improvement and the impact of policy stock sales - Correlation analysis between capital efficiency and EPS growth - Impact of policy stock sales on ROE improvement - Calculation and comparison of EPS growth rate

4.4. Impact of varying levels of accumulation of research memos on report generation

We examined how the characteristics of reports generated by Expert Agents evolve as the number of accumulated memos increases. We took a similar manner used by Babaei and Giudici (2024). Specifically, we compared the following cases: Expert Agents holding the first generated 25 research memos versus Novice Agents with no memos; Expert Agents with the first 50 memos versus those holding 25 memos; and Expert Agents possessing all 70 to 75 memos versus those with 50 memos.

Table 8 presents the evaluation results for each of the aforementioned pairs of agents under the eight evaluation criteria, as assessed by GPT-4.1. Our findings confirm that as the number of memos increases, Expert Agents demonstrate consistent improvements in the “unique perspectives” and “profound insights” criteria, while no consistent changes are observed in the other criteria. This indicates that accumulating research memos covering a wider range of topics provides Expert Agents with more examples to draw upon, thereby enabling them to generate more unique perspectives and profound insights in their analyses.

4.5. Effect of including research memos in the prompt for GPT during evaluation

In the above experiments, when using GPT-4.1 for evaluation, we did not provide the research memos to GPT. In this section, we examine whether providing the research memos to GPT can bring its assessment results closer to those of human analysts.

To measure the difference between GPT's evaluation and that of human analysts, we first convert the evaluation results to scores in the following manner. For each evaluation criterion of each sample, a score of -1 is assigned if the Novice Agent is judged superior, $+1$ if the Expert Agent is judged superior, and 0 if they are rated as equal. This scoring method ensures that the difference between the Novice

and Expert Agents is 2, while the difference between either agent and the equal rating is 1. In our experiment, each sample was evaluated by three analysts and six GPT evaluators. The average score across each type of evaluators was taken as the score for each criterion in each sample. Thus, for sample (i) and criterion (j), we obtain the human analyst score y_{ij}^h and the LLM score y_{ij}^l . For the 50 samples evaluated by human analysts, we use the RMSE to compute the difference between human and GPT evaluations. RMSE is calculated as follows:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^C (y_{ij}^h - y_{ij}^l)^2}$$

where $N = 50$ and $C = 8$.

First, Table 9 presents GPT's evaluation results both without and with research memos included in prompts. For five out of the eight evaluation criteria, both approaches yield the same judgment as to whether the Novice or Expert Agent is superior, while the other three show discrepancies. This demonstrates that providing research memos affects GPT's judgement.

Next, Table 10 presents the differences among the evaluations of human analysts, GPT without memos, and GPT with memos. The difference between the GPT without memos and human analysts is smaller than that between GPT with memos and human analysts. Therefore, in this study, it is more appropriate to use GPT without research memos for evaluation purposes.

4.6. Impact of different prompts on report generation

In addition to research memos, we investigated whether different prompt designs affect the quality of generated reports. To this end, we conducted a series of experiments using three distinct prompts to instruct GPT-4o to conduct company analysis. As shown in Table 11, Prompt A simply provides the company name and its earnings releases, asking GPT to analyze a specific research topic. Prompt B assigns GPT-4o the role of a senior fundamental analyst, while Prompt C further

Table 8

Performance comparison of reports generated by Expert Agents possessing different number of research memos. Evaluated by six LLM evaluators for all the 100 topics.

	Novice vs. Expert (25 memos)			Expert (25 memos) vs. (50 memos)			Expert (50 memos) vs. (all memos)		
	Novice	25 memos	Equal	25 memos	50 memos	Equal	50 memos	all memos	Equal
1.1 Quantitative Analysis	49%	51%	0%	45%	55%	0%	52%	48%	0%
1.2 Qualitative Analysis	50%	50%	0%	57%	44%	0%	39%	60%	0%
2.1 Short-Term Factors	52%	49%	0%	45%	55%	0%	41%	59%	0%
2.2 Long-Term Factors	52%	48%	0%	45%	55%	0%	53%	47%	0%
3.1 Direct Relevance	51%	49%	0%	46%	54%	0%	58%	42%	0%
3.2 Unique Perspectives	39%	61%	0%	50%	50%	0%	47%	53%	0%
4.1 Basic Information	46%	54%	0%	52%	48%	0%	53%	47%	0%
4.2 Profound Insights	47%	53%	0%	49%	51%	0%	48%	52%	0%

More research memos consistently lead to higher performance in “unique perspectives” and “profound insights”.

Table 9

Comparison of evaluation by LLMs with and without research memos included in prompts. Bold values indicate that both evaluation yield the same judgment as to whether the Novice or Expert Agent is superior.

	Evaluation by LLM w/o memos			Evaluation by LLM with memos		
	Novice	Expert	Equal	Novice	Expert	Equal
1.1 Quantitative Analysis	54%	46%	0%	60%	40%	0%
1.2 Qualitative Analysis	47%	54%	0%	53%	47%	0%
2.1 Short-Term Factors	52%	49%	0%	57%	44%	0%
2.2 Long-Term Factors	48%	51%	1%	56%	45%	0%
3.1 Direct Relevance	48%	52%	0%	54%	46%	0%
3.2 Unique Perspectives	40%	60%	0%	48%	52%	0%
4.1 Basic Information	51%	49%	0%	64%	36%	0%
4.2 Profound Insights	46%	54%	0%	47%	53%	0%

Table 10

Distance (RMSE) between evaluations by analysts and GPT-4.1.

	Analyst	GPT w/o memos	GPT w/ memos
Analyst	–	2.42	2.57
GPT w/o memos	–	–	1.68
GPT w/ memos	–	–	–

instructs GPT-4o to perform an in-depth analysis. Prompt C corresponds to the prompt used by the Novice Agent in our study. The research topics used in this experiment are the 100 topics described in Section 3. For each research topic, we input the 2024 earnings releases of the corresponding target companies into the prompt.

We evaluated the reports generated by each prompt using the method described in Section 2. Specifically, we asked GPT-4.1 to compare Prompt B versus Prompt A, and Prompt C versus Prompt B. Table 12 presents the evaluation results. Bold values indicate evaluation criteria for which the two prompts in each pair resulted in a substantial performance gap.

It can be observed that assigning GPT-4o the role of an analyst consistently improved or maintained its performance across all criteria, with particularly notable improvements in generating “long-term factors” and “unique perspectives”. Further instructing the model to conduct an in-depth analysis led to a decrease in “quantitative analysis” while an increase in “qualitative analysis”, along with substantial improvements in “unique perspectives”, “basic information”, and “profound insights”.

These results indicate that, in the context of company analysis, both assigning a role and specifying the type of analysis can influence the output generated by GPT. With respect to our study’s focus on enhancing performance in “unique perspectives” and “profound insights”, both of these prompting strategies contribute positively. This result aligns with the findings of Zhao et al. (2025), which reported that the use of prompts and role-play settings of LLMs significantly influence creativity.

5. Conclusion

To enhance the capability of LLMs in assisting humans in fundamental analysis, we conducted the following study:

1. In fundamental analysis practices, it is necessary to selectively employ relevant information to analyze various research topics for multiple companies within the same sector. Based on this domain-specific characteristic, we developed a novel “Autonomous Fundamental Analysis System (AutoFAS)”, enabling LLM agents to autonomously conduct comprehensive analyses on various research topics.
2. Inspired by the characteristics of human analysts, we proposed and validated a novel direction to enhance the fundamental analysis capabilities of LLMs. Within the AutoFAS, we let LLM agents autonomously conduct comprehensive analyses of companies within specific sector, accumulating understandings of various aspects of these companies. This enables the agents to naturally provide more analyses on long-term factors and offer more unique and profound insights for new research topics. This characteristic exhibited by LLMs aligns with human thought processes.

CRedit authorship contribution statement

Tao Xu: Conceptualization, Formal analysis, Methodology, Software, Validation, Investigation, Writing – original draft. **Zhe Piao:** Conceptualization, Investigation, Data curation. **Tadashi Mukai:** Conceptualization, Investigation. **Yuri Murayama:** Conceptualization, Writing – review & editing. **Kiyoshi Izumi:** Conceptualization, Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Table 11
Prompts with different key phrases.

Prompt A	Prompt B	Prompt C
Presented below are the earnings releases of {company name}. Please analyze the topic {research topic} based on this information. {earnings releases}	You are a senior fundamental analyst working at a financial institution. Presented below are the earnings releases of {company name}. Please analyze the topic {research topic} based on this information. {earnings releases}	You are a senior fundamental analyst working at a financial institution. Presented below are the earnings releases of {company name}. Please analyze the topic {research topic} based on this information. Please conduct an in-depth analysis. {earnings releases}

Table 12
Performance comparison of reports generated with different prompts. Bold values indicate substantial performance gaps between the two prompts in each pair.

	Prompt A vs. B			Prompt B vs. C		
	Prompt A	Prompt B	Equal	Prompt B	Prompt C	Equal
1.1 Quantitative Analysis	37%	43%	20%	47%	23%	30%
1.2 Qualitative Analysis	47%	53%	0%	40%	60%	0%
2.1 Short-Term Factors	50%	50%	0%	50%	50%	0%
2.2 Long-Term Factors	33%	67%	0%	50%	50%	0%
3.1 Direct Relevance	40%	40%	20%	33%	40%	27%
3.2 Unique Perspectives	30%	70%	0%	37%	63%	0%
4.1 Basic Information	43%	50%	7%	30%	57%	13%
4.2 Profound Insights	50%	50%	0%	40%	60%	0%

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Data availability

The authors do not have permission to share data.

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